
2026 BANKING ARCHITECTURE: A FRAMEWORK FOR OPERATIONAL RESILIENCE

WHITEPAPER

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June 2026

ABSTRACT

The 2026 banking landscape is defined by three forces moving in parallel. The Digital Operational Resilience Act [1] in the EU, the Federal Reserve’s SR 21-14 [2] and the OCC’s Heightened Standards [3] in the United States, and APRA CPS 230 [4], MAS Technology Risk Management [5], and HKMA C-RAF [6] in Asia-Pacific have together elevated legacy cryptographic debt and supply-chain fragility from operational hygiene concerns into board-accountable regulatory liabilities. The November 2026 SWIFT MT/MX cut-over renders MT103-based translation strategies structurally obsolete — according to the RedCompass Labs ISO 20022 Readiness Index 2025, 44 percent of surveyed banks are off-track for the deadline [7]. And the rise of multi-rail liquidity, captured in the BIS Project Agorá workstream [8] and the BIS analysis of stablecoins versus tokenised deposits [9], has shifted the competitive question from which bank to which rail, under what policy, and against what liquidity buffer.

This whitepaper presents a three-pillar architectural framework — the *Resilience Trinity* — composed of cryptographic stewardship (with explicit post-quantum migration semantics aligned to NIST FIPS 203/204/205 [10–12] and NIST IR 8547 [13]), an ISO 20022 canonical data substrate, and rail-agnostic orchestration governed by policy-as-code. It sets out a sequential three-phase implementation roadmap aligned with the 2026 DORA reporting cycle, the November 2026 SWIFT cut-over, and the Basel intraday-liquidity monitoring framework [14]. The framework is positioned alongside, not against, vendor core-banking modernisation and API-first open-banking approaches; the decision matrix in Section 7 documents where it complements them and where it competes.

Keywords Banking architecture · DORA · ISO 20022 · Programmable liquidity · Tokenised deposits · Multi-rail orchestration · Policy-as-code · Operational resilience

1 Executive Summary

The 2026 banking landscape is defined by three forces moving in parallel.

Regulatory convergence on operational resilience. DORA [1] entered application in January 2025 in the EU and assigns explicit board accountability for ICT risk under Article 5. The UK regime, in force from March 2025, combines PRA SS1/21 [15] and FCA PS21/3 [16] on impact tolerances and important business services. In the United States, the Federal Reserve’s SR 21-14 [2] codifies sound practices for operational resilience, layered above the OCC’s heightened standards for large national banks [3] and the FFIEC IT Examination Handbook [17]. In Asia-Pacific, MAS Technology Risk Management Guidelines [5], HKMA C-RAF 2.0 [6], and APRA CPS 230 [4] converge on the same posture. The Basel Committee’s Principles for Operational Resilience [18] provide a common international floor. Stagnant Argon2id parameters [19], un-peppered hashes, and unvetted C foreign-function-interface dependencies are no longer technical-debt line items in any of these jurisdictions; they are findings waiting to be written.

Structural data shift in cross-border payments. The November 2026 SWIFT MT/MX cut-over renders MT103-based translation strategies obsolete. Banks still emitting unstructured remittance and address data will be surcharged on every message and cut off from MX-only correspondents. The BIS Committee on Payments and Market Infrastructures (CPMI) ISO 20022 harmonisation requirements [20] are a precondition for the G20 cross-border payments programme targets on cost, speed, transparency, and access [21]. According to the RedCompass Labs ISO 20022 Readiness Index 2025 [7], which surveyed cross-border payment leaders at 200 Tier-1 and Tier-2 banks across Europe, North America, and Asia-Pacific, 44 percent of respondents report being off-track for the November 2026 deadline. The McKinsey 2024 Global Payments Report [22] sizes the addressable cross-border B2B flow at over USD 150 trillion annually, and the Capgemini World Payments Report 2025 [23] documents the parallel acceleration of instant-payment rail adoption.

Multi-rail liquidity and the orchestration question. The rise of multi-rail liquidity — SWIFT CBPR+ [24], PSD3/A2A (FedNow [25], SEPA Instant [26], RTP, and the major instant payment systems in Asia and Latin America), and tokenised deposits — has shifted the competitive question from *which bank do we use* to *which rail does this payment go down, and under what policy*. The BIS Project Agorá workstream [8] and the BIS analysis of stablecoins versus tokenised deposits [9] document why the orchestration layer, not the rail, is where margin lives in 2026.

Scope and method. This whitepaper sets out a three-pillar architectural framework — the Resilience Trinity — to address these three forces with a single, coherent operating model. The framing builds on the established payments-economics literature [27] on why payment system design has first-order effects on financial-system function, and on the central-bank-digital-money lineage developed in the BIS Quarterly Review [28, 29] and the IMF FinTech notes [30]. The framework is presented alongside, not against, vendor core-banking modernisation programmes (such as those built around Temenos, Mambu, Finastra, or FIS) and pure API-first open-banking platform models; a comparative posture matrix is documented in Section 7. The framework’s components have been independently exercised in published wholesale-banking pilots — J.P. Morgan’s Onyx Digital Assets [31] and HSBC’s Orion platform [32] are the most cited public examples for tokenised deposits and digital-asset settlement. The framework synthesises these lines of practice rather than introducing a parallel one.

2 The Resilience Trinity

We propose a three-pillar framework for modernising the core banking stack: **hardened security**, **canonical data**, and **multi-rail orchestration**. Each pillar addresses one of the three converging pressures introduced above. Each is independently valuable; together they compose an end-to-end posture that survives the 2026/2027 regulatory cycle and the parallel post-quantum cryptography transition documented in NIST IR 8547 [13].

- **Pillar I — Cryptographic Stewardship.** Move beyond C-based foreign-function-interface to memory-safe cryptographic frameworks with multi-algorithm dispatch, hardware-security-module (HSM)-interlocked peppering, `verify_and_upgrade` semantics, and a defined migration path to NIST-standardised post-quantum primitives (FIPS 203/204/205 [10–12]).
- **Pillar II — The ISO 20022 Data Substrate.** Adopt an ISO-first canonical schema across every API contract, validation gate, and downstream consumer. Translate MT up to MX once at ingress; never carry MT downstream. Enforce a stricter-than-CBPR+ message profile so structured purpose, remittance, and regulatory data carry through correspondent legs without truncation.
- **Pillar III — Rail-Agnostic Orchestration.** Move orchestration out of the model and into a policy-as-code engine that routes payments based on corridor, ticket size, settlement risk, and counterparty relationship — with the treasury agent acting only within the bounds the policy defines, and with intraday-liquidity buffers documented per rail per corridor in line with BCBS 248 [14].

The framework is deliberately modular. A bank constrained by a 2026 DORA reporting deadline can sequence the pillars to match its compliance pressure points; a bank competing for treasury mandate share can sequence them to match its commercial road-map. The dependencies run one way: Pillar II is harder without Pillar I (one cannot enforce schema integrity on a control plane whose authentication primitives one does not trust), and Pillar III is harder without Pillar II (one cannot route on machine-readable purpose codes one does not have).

Position relative to alternative approaches. The Resilience Trinity is one of several available frameworks for 2026 banking modernisation. Vendor-led core-banking modernisation (Temenos, Mambu, Finastra, FIS) addresses much of Pillar II’s data canonicalisation by adoption; an API-first open-banking platform model addresses parts of Pillars II and III but does not, by itself, remediate cryptographic debt or enforce policy-as-code at the orchestration layer; a

CBDC-rail thesis addresses settlement but does not address the messaging or control-plane questions. The matrix in Section 7 documents where the Resilience Trinity complements each of these and where it competes.

3 Pillar I — Cryptographic Stewardship

The Foundation

In the era of DORA, SR 21-14, CPS 230, and the parallel post-quantum transition signalled by NIST IR 8547 [13] and the NSA's CNSA 2.0 suite [33], “deploy-and-forget” password hashing is a systemic liability. Cryptographic rot — stagnant Argon2id parameters relative to RFC 9106 floors [19], un-peppered hashes, supply-chain-exposed C foreign-function-interface (FFI) — is no longer a technical debt line; it is a regulatory finding waiting to be written under any of the regimes cited in the executive summary.

Threat Model

Four attacker classes drive the engineering posture.

1. **Offline credential cracking after a database breach.** The adversary obtains the password store and runs GPU- or ASIC-accelerated dictionary attacks against the stored hashes. Defence requires current memory-hard parameters (RFC 9106 Argon2id) and HSM-resident peppering so a database breach alone yields nothing crackable.
2. **Supply-chain compromise of cryptographic primitives.** The adversary plants a backdoor or undisclosed weakness in a widely deployed C cryptographic library reached through FFI. NIST SP 800-218 (SSDF) [34] treats supply-chain provenance as a first-class control; ENISA's 2024 threat landscape [35] repeatedly highlights cryptographic-library compromise as a recurring vector. Defence requires memory-safe, audit-able, reproducible-build cryptographic primitives with explicitly enumerated transitive dependencies.
3. **Harvest-now, decrypt-later (HNDL) attack on long-lived secrets.** The adversary captures encrypted traffic or persisted ciphertext today on the assumption that a cryptographically relevant quantum computer (CRQC) becomes available later in the document's protection horizon. Defence requires a published migration plan to NIST FIPS 203 (ML-KEM) [10], FIPS 204 (ML-DSA) [11], and FIPS 205 (SLH-DSA) [12], with timelines aligned to NIST IR 8547 (which calls for RSA and ECC retirement for key establishment by 2035) and to NSA CNSA 2.0 (which mandates ML-KEM and ML-DSA for national security systems by January 2027).
4. **Insider misuse of administrative credentials.** The adversary is a privileged user with key-store access. Defence requires HSM- or KMS-mediated key release, request-scoped pepper use (no string concatenation), and audit-grade key-rotation telemetry.

The Thesis (classical)

For the classical password and authentication estate, three engineering properties make the pillar work.

1. **Multi-algorithm dispatch.** The verifier inspects the PHC string at parse time, routes to the correct algorithm (PBKDF2, scrypt, Argon2id), validates the password, and — if the stored algorithm or parameter set is below the current RFC 9106 floor [19] — re-hashes silently with the current standard. User-visible behaviour is unchanged. The authentication estate strengthens by one record on every successful login.
2. **HSM-interlocked peppering.** The pepper is sourced from an HSM or KMS at request scope, consumed via the cryptographic library's secret parameter (*not* via string concatenation), and dropped from memory after use. A database breach alone yields nothing crackable; a key-store breach alone yields a secret with no usable target. The system loses confidentiality only if both stores fail together.
3. **Memory-safe, reproducible-build supply chain.** No unaudited C FFI; transitive dependencies enumerated; cryptographic primitives memory-safe at the language level and audit-able at the source level. Owasp's Password Storage guidance [36] treats algorithm and parameter currency as a live operational obligation, not a one-time choice.

Post-Quantum Migration

For the long-lived secrets and the signing estate, the pillar additionally requires a published PQC migration plan.

- **Key encapsulation.** Migrate the TLS, S/MIME, and inter-service key-exchange estate to FIPS 203 ML-KEM [10], in hybrid mode (classical + PQC) during the transition period defined by NIST IR 8547 [13].
- **Digital signatures.** Migrate the code-signing, software-attestation, and document-signing estate to FIPS 204 ML-DSA [11], with FIPS 205 SLH-DSA [12] as the hash-based fall-back where stateful or conservative-assumption signatures are required.
- **Inventory and crypto-agility.** The remediation is impossible without a cryptographic-asset inventory mapping every primitive, parameter, and lifetime to a business process. Crypto-agility — the ability to swap a primitive without redeploying the business code that uses it — is the structural prerequisite. The pure-Rust posture above is one path; HSM-vendor-supplied PQC libraries are another. The framework is agnostic on choice but firm on requirement.

Choosing the Crypto-Agility Implementation Path

The framework is firm on the requirement of crypto-agility but agnostic on the implementation choice. The decision between (a) a memory-safe-language cryptographic substrate (commonly Rust, Go, or memory-safe C dialects) and (b) HSM-vendor-supplied PQC libraries is a function of four operational factors. The matrix below provides an ARB-actionable starting point; institutions should refine the weightings against their own threat model and operating envelope. Foundational analyses of the PQC transition [37] and of memory-hard hashing [38] provide the underlying technical justification.

Factor 1 — System topology. If cryptographic primitives are reached from many call sites distributed across services and languages, an HSM-vendor library — exposed over a uniform PKCS#11 or equivalent interface — minimises duplication of cryptographic state. If primitives are concentrated in one or two long-lived authentication services, a memory-safe in-process substrate gives finer control over parameter currency and secret lifetimes.

Factor 2 — Existing vendor relationship. If the institution already operates an HSM estate under contractual coverage that includes PQC primitives within the NIST IR 8547 [13] timeline, the marginal cost of the HSM-vendor path is low. If the existing HSM estate lacks committed PQC coverage, the institution will incur a parallel cost regardless of choice; the memory-safe path then competes on its own merits.

Factor 3 — Timeline pressure. If the NSA CNSA 2.0 January 2027 mandate [33] bites (national-security workloads, US federal banking lines), the HSM-vendor path is typically faster to credible deployment because the certification work is the vendor's. If the binding timeline is the broader NIST IR 8547 horizon to 2035, the memory-safe path has time to mature and offers stronger supply-chain provenance under NIST SP 800-218 [34].

Factor 4 — In-house engineering capability. The memory-safe path requires sustained ownership of cryptographic-substrate code; in-house engineering depth must be commensurate. The HSM-vendor path transfers that obligation but introduces a vendor concentration risk that must be documented in the programme risk register (Section 7).

The decision is rarely binary. A common pattern is the HSM-vendor path for long-lived signing primitives (where vendor certification value is high) combined with a memory-safe in-process substrate for the password and session-key estate (where call-site density and parameter currency dominate). The framework accepts either or both, provided the resulting estate is documented, monitored for parameter currency, and exhibits a credible PQC migration trajectory.

Regulatory Mapping

Article 5 of DORA [1] assigns board accountability for ICT risk management. Article 7 requires identification and protection of ICT assets supporting critical functions; Article 9 requires cryptographic-key management. The OCC's heightened standards [3] require an enterprise risk-management framework that explicitly addresses information-technology risk. The Fed's SR 21-14 [2] flags governance, third-party risk management, and surveillance as sound-practice domains. MAS TRM [5], HKMA C-RAF 2.0 [6], and APRA CPS 230 [4] converge on the same expectations from the APAC side. A cryptographic-stewardship posture built on the threat model and the classical-plus-PQC theses above gives the board a defensible answer to all five jurisdictions: every password store is audited; every algorithm and parameter is current; every pepper is HSM-resident; every credential read is logged with cryptographic provenance; every long-lived secret has a documented PQC migration path.

Companion technical write-up

The engineering detail behind the classical thesis — including `verify_and_upgrade` dispatch semantics, HSM-interlock sequence flows, and a parameter-currency telemetry pattern — is set out in the companion write-up [39]. The companion material is illustrative, not load-bearing for the framework: any memory-safe cryptographic library

with multi-algorithm dispatch, RFC 9106 parameter currency, HSM-interlocked peppering, and a documented PQC migration plan satisfies the pillar.

4 Pillar II — The ISO 20022 Data Substrate

The Language

With the November 2026 SWIFT MT/MX cut-over, ISO 20022 [40] is the non-negotiable data substrate. It is not a migration project; it is the wiring for agentic treasury.

The structural problem with the predecessor format is data. MT103 carries 35 characters of unstructured remittance in field 70 and a free-text address in field 50K. pacs.008 carries <RmtInf> with structured creditor reference, <PstlAdr> with street, post code, town, and country code as discrete elements, and <RgltryRptg> for jurisdiction-specific obligations. The 2024 CBPR+ uplift [24] turned what were once “may” fields into “must” fields. Banks that translate down to MT103 lose the data they need to satisfy FATF Recommendation 16 on originator and beneficiary information [41].

Without structured <Purp> codes, structured <PstlAdr> fields, and structured <RmtInf> remittance, a treasury agent has nothing to reason over. It has prose.

The Empirical Picture

According to the RedCompass Labs ISO 20022 Readiness Index 2025 [7], 44 percent of surveyed banks are off-track for the November 2026 cut-over. The McKinsey 2024 Global Payments Report [22] sizes the cross-border B2B payment flow these banks need to address in the new format at over USD 150 trillion annually. The Capgemini World Payments Report 2025 [23] documents the parallel acceleration of instant-payment volumes, all of which run on ISO 20022 from day one. The BIS Triennial Central Bank Survey [42] provides the most-cited reference for the FX leg of these flows. The pressure is therefore not one event in November 2026; it is the structural fact that every adjacent rail and every adjacent regulatory regime now assumes the same schema.

The Thesis

Adopt an ISO-first canonical schema across every API contract, validation gate, and downstream consumer. Reject on parse, not on settlement. Stop translating MX down to MT at the edge — translate MT *up* to MX once and discard the MT.

In practical terms, an agentic treasury automatically optimises intraday liquidity positioning by reconciling structured <Purp> codes and remittance data against real-time funding needs — moving cash, drawing on credit lines, or holding back execution without a human in the loop, all within bounds defined by the operational-resilience regime. The MX message is the nerve impulse. The treasury control plane is the spinal cord. SR 11-7 model-risk governance [43] and the operational-resilience regimes cited in Section 1 sit on top as the central nervous system. Strip the MX away and the agents go blind.

Multi-Jurisdictional Reality

The schema is global; the rails it lands on are not. FedNow [25] ships pacs.008 / pacs.002 on a 24×7 US domestic rail. SEPA Instant [26] carries the same messages on the European 10-second rail. In Latin America, Brazil’s PIX [44] runs natively on ISO 20022 from launch and has become the most-cited reference architecture for state-sponsored instant payments; Mexico’s CoDi [45] layers on top of the SPEI ISO 20022 profile; Argentina’s Transferencias 3.0 [46] brings interoperable QR-initiated instant payments under the same canonical messaging shape. The major APAC instant payment systems — UPI in India, FAST in Singapore, FPS in Hong Kong, NPP in Australia — all converge on ISO 20022 messaging, often with locally extended schemas. The treasury workstation reads a pain.001 once and routes per corridor. The CIB that wins on cross-border in 2026 enforces a stricter message profile than CBPR+ requires — and rejects on parse, not on settlement.

Operational Consequences

Three consequences follow directly from the ISO-first posture.

- **Routing is data-driven.** The treasury workstation reads a pain.001 once and decides the rail per payment based on corridor, ticket size, cut-off, and counterparty relationship — without re-mapping the message.
- **Remittance data survives the hop.** Structured remittance fields (<RmtInf><Strd>) carry through correspondent legs without truncation. Auto-reconciliation rates climb because the data is no longer lost at the rail boundary.
- **Sanctions screening becomes auditable.** Structured <Dbtr> / <Cdtr> / <DbtrAgt> / <CdtrAgt> fields with Legal Entity Identifier (LEI) references [47] replace free-text name screening. Hit rates fall. Investigation queues shrink.

Companion technical write-up

The companion technical write-up [48] walks through a worked pain.001 fragment with the structured-address fields populated and documents the validation profile a CIB ingress should enforce. The write-up is illustrative; any ISO-first canonical schema that rejects on parse, carries structured remittance end-to-end, and binds sanctions screening to LEI references satisfies the pillar.

5 Pillar III — Rail-Agnostic Orchestration

The Execution

Treasury in 2026 is no longer about picking a bank — it is about picking a rail. SWIFT CBPR+ [24], PSD3/A2A, and tokenised deposits are commodity execution venues. Success lies in the orchestration layer that binds them — and in keeping that layer *outside* the agent so model risk, audit, and operational-resilience accountability remain enforceable.

The Two Tokenised-Settlement Models

The wholesale settlement leg now splits into two structurally different token models, the trade-off between which is captured in the BIS analysis of stablecoins versus tokenised deposits [9].

Bank-issued tokenised deposits. A direct claim on the issuing bank on a permissioned ledger. J.P. Morgan’s Onyx Digital Assets platform [31] and HSBC’s Orion platform [32] are the most-cited Tier-1 examples; the BIS Project Agorá workstream [8] extends the model to multi-bank cross-border settlement with central-bank-money interoperability. Settlement is near T+0 inside the network. Compliance is the issuing bank’s responsibility; the rail is fully regulated, fully traceable, and constrained to participants the issuer has on-boarded.

Integrated regulated stablecoin rails. A claim on the reserve, audited under MiCA [49] in the EU, under the Bank of England’s discussion-paper regime for systemic stablecoins [50] in the UK, and under the analogous regional frameworks elsewhere. Compliance is shared between issuer, on-ramp, and off-ramp. The two models are not competing; they are stacked. A CIB cross-border product in 2026 typically uses bank-issued tokenised deposits for the in-network leg and a regulated stablecoin for the corridor where the in-network rail terminates. The corporate sees one ISO 20022 payment. The settlement story underneath is multi-token.

The Liquidity Buffer Question

The board-level question is the same one operational-risk committees have been asking since the first programmable-liquidity pilots: *who carries the credit exposure on the token, and for how long?* The answer is jurisdiction-specific but has a documented quantitative envelope.

- **LCR treatment.** Under the Basel III Liquidity Coverage Ratio [51], tokenised deposits on a permissioned ledger inherit the High-Quality Liquid Asset (HQLA) classification of the underlying deposit. Regulated stablecoins are not HQLA; reserves backing them may include HQLA, but a treasury position in the stablecoin itself is treated as a counterparty exposure, not as a liquidity buffer.
- **NSFR treatment.** The Net Stable Funding Ratio [52] weights funding stability over a one-year horizon. Tokenised deposits roll up to the issuing bank’s funding profile; stablecoin balances do not. A treasury team relying on stablecoin float for medium-term funding under-states its required stable funding.
- **Intraday liquidity.** The BCBS intraday-liquidity monitoring framework [14] requires daily maximum intraday liquidity usage to be reported per currency. A multi-rail control plane must aggregate the maximum 15-minute net outflow per rail per corridor and hold a documented buffer. *Practitioner heuristic, not regulatory minimum.*

A working starting point used by some treasury control-plane implementations is 125 percent of the rolling 30-day per-corridor peak, refined as longitudinal data accumulates. BCBS 248 itself defines the monitoring metrics and reporting cadence, not a quantitative buffer floor; treasury teams should size buffers against their own intraday stress test results and supervisory expectation, taking the heuristic only as a starting point for calibration.

A treasury team that does not document the answer per rail per corridor is carrying unmeasured credit risk on its balance sheet. The orchestration layer is what makes the documentation tractable: every routing decision is a function of policy state (rail, corridor, ticket size, counterparty, time of day) and writes an audit record sufficient to reproduce the LCR/NSFR/intraday-liquidity attribution after the fact.

The Thesis

Move orchestration out of the model and into a policy-as-code engine that routes payments based on corridor, ticket size, settlement risk, and counterparty relationship — with the agent acting only within the bounds the policy defines, and with intraday-liquidity buffers documented per rail per corridor in line with BCBS 248 [14].

The architecture is one ingress, one canonical message, three rails, one settlement reconciliation, and one liquidity attribution.

1. Corporate Enterprise Resource Planning (ERP) emits a pain.001 ISO 20022 message.
2. Bank ingress schema-validates against the bank's stricter-than-CBPR+ profile; rejects on parse.
3. The policy-as-code orchestrator routes per corridor, ticket size, and counterparty relationship:
 - High-value cross-border → SWIFT CBPR+ (pacs.008).
 - Domestic instant within cut-off → A2A / Open Finance (PSD3 / FedNow / SEPA Instant / RTP).
 - In-network tokenised corridor → Tokenised deposit on permissioned ledger (Onyx / Orion / Agorá class).
4. Settlement status returns as pacs.002, regardless of rail.
5. Auto-reconciliation consumes structured <RmtInf> end-to-end.
6. LCR / NSFR / intraday-liquidity attribution writes per rail per corridor to the supervisory reporting line.

Worked Example: a Policy-as-Code Routing Rule Set

The routing layer is policy-as-code by design. The shape of the policy matters more than the choice of engine (Open Policy Agent / Rego, Cedar, an internal DSL, or a tabular rules engine all suffice provided the policy is auditable, version-controlled, and separable from the agent). A minimum-viable rule set has four rules: the in-network tokenised corridor, the high-value cross-border corridor, the domestic-instant corridor, and the explicit overnight-correspondent fallback. A schema-level sketch is below.

```
# Policy: Bank ABC multi-rail routing
# Eval order: first-match-wins.
# Each rule writes both the route decision and the
# Basel attribution metadata the supervisor expects.

rules:

- id: in_network_tokenised
  when:
    counterparty.is_onboarded_tokenised_participant: true
    iso20022.purpose_code: [TRAD, INTC, SUPP]
    amount.currency:      [USD, EUR, GBP]
  route:
    rail:                  tokenised_deposit
    settlement_finality: T+0
  attribution:
    hqla_class:            Level_1
    credit_exposure:       issuing_bank
    bcbs_248_metric:       intraday_max_15min
```

```

- id: high_value_cross_border
  when:
    amount.eur_equivalent: { gt: 1_000_000 }
    corridor.requires_cbpr_plus: true
  route:
    rail: swift_cbpr_plus
    schema: pacs.008.001.10
  attribution:
    hqla_class: Level_1
    credit_exposure: correspondent_bank
    bcbs_248_metric: intraday_max_15min

- id: domestic_instant_a2a
  when:
    corridor.is_domestic: true
    cutoff.within_window: true
    amount.eur_equivalent: { lt: 100_000 }
  route:
    rail_by_jurisdiction:
      US: fednow
      EUR: sepa_instant
      UK: faster_payments
      BR: pix
      MX: codi_spei
    max_latency_ms: 10_000
  attribution:
    hqla_class: Level_1
    credit_exposure: domestic_clearing
    bcbs_248_metric: intraday_max_15min

- id: fallback_overnight_correspondent
  when: { match: any }
  route:
    rail: swift_cbpr_plus
    schema: pacs.008.001.10
  attribution:
    hqla_class: Level_1
    credit_exposure: correspondent_bank
    bcbs_248_metric: intraday_max_15min

```

Three properties make the example load-bearing rather than illustrative. First, every rule writes the BCBS 248 attribution metadata [14] alongside the route decision, so intraday-liquidity supervisory reporting becomes a join, not a separate pipeline. Second, the agent's role is bounded to populating the when-clause inputs (corridor classification, counterparty onboarding status, cut-off-window check); the route decision is the policy engine's, not the model's. Third, the fallback rule is explicit, so an agent failure or an unrecognised corridor produces a deterministic correspondent-banking route rather than an undefined outcome.

Governance of the Agentic Layer

SR 11-7 [43] is the foundational text on model risk management and remains the supervisory reference for any model embedded in the orchestration layer. It pre-dates the generation of language-model-based agents now being considered for treasury workflows, so it is not, on its own, sufficient guidance for governing an agentic treasury estate. Practitioners building this layer in 2026 should treat SR 11-7 as the floor and overlay current AI-specific guidance.

International. The Financial Stability Board's 2017 report on artificial intelligence and machine learning in financial services [53] sets out the systemic-stability lens against which any agentic treasury deployment will be assessed. The BIS Financial Stability Institute's review of AI regulation [54] catalogues the supervisory tools available across jurisdictions; the catalogue informs which controls are likely to be requested in supervisory dialogue.

Europe. The EU AI Act [55] entered into force in 2024 and applies horizontally. A treasury agent that materially affects payment outcomes will, in most institutional deployments, fall within the Act's high-risk classification, with the documentation, transparency, and human-oversight obligations that classification carries. ESMA's 2023 statement on AI in retail investment services [56] is the closest sectoral analogue and signals supervisory expectations on disclosure and oversight.

Practical implication. The orchestration layer should be designed such that the agent never holds the routing decision; the policy-as-code engine holds it. The agent's role is to propose, within bounded action sets, against policy state. The audit trail records both the agent proposal and the policy decision separately; any deviation is, by construction, a policy-as-code event, not a model event. This separation is what makes the SR 11-7 + AI-Act compliance posture tractable: the bounded surface area on which model risk is non-zero is small, documented, and demonstrable to a supervisor.

Companion technical write-up

The corridor-level mechanics — including the message-routing matrix, the per-rail credit-exposure tables, and the integration patterns with bank-issued tokenised-deposit platforms — are documented in the companion write-up [57]. The companion material is illustrative; any policy-as-code orchestration layer that consumes ISO 20022 natively, routes per documented credit-exposure profile, and writes the Basel intraday-liquidity attribution per rail per corridor satisfies the pillar.

6 Architectural Implementation Roadmap

Three sequential phases. Each is independently valuable; together they compose the Resilience Trinity end-to-end. The roadmap aligns with the 2026 DORA reporting cycle, the November 2026 SWIFT MT/MX cut-over, the 2027 deepening of programmable-liquidity adoption signalled by BIS Project Agorá [8], and the 2027/2035 PQC migration boundaries set by NSA CNSA 2.0 [33] and NIST IR 8547 [13].

Phase 1 — Audit and Secure (months 0–9)

Remediate cryptographic rot using memory-safe primitives to meet DORA, SR 21-14, CPS 230, MAS TRM, and HKMA C-RAF resilience mandates.

Technical work.

- Inventory every password store, key-derivation-function parameter set, and cryptographic library — including indirect C dependencies behind FFI layers.
- Migrate to a memory-safe cryptographic framework with `verify_and_upgrade` dispatch, HSM-interlocked peppering, and audit-grade key-rotation telemetry.
- Publish a PQC migration plan covering FIPS 203 / 204 / 205 [10–12], aligned to NIST IR 8547 and NSA CNSA 2.0 timelines.
- Document the migration as a board-accountable change under DORA Article 5 [1] and as a NIST SP 800-218 supply-chain control [34].

Organisational change management.

- Stakeholder mapping: CISO, ICT risk function, internal audit, ARB chair as accountable executive; Pillar I sponsor.
- Vendor dependency risk: avoid single-vendor lock-in for the cryptographic substrate; require source availability and reproducible builds for any primitive on the critical path.
- Staff retraining: legacy C / Java cryptographic-platform teams reskill to memory-safe languages or to a wrapped-FFI pattern; security-architecture team upskill on PQC primitives and crypto-agility patterns.
- Programme-governance gates: ARB sign-off on the cryptographic-asset inventory; CISO sign-off on the HSM peppering design; internal-audit attestation on the PQC migration plan.

Exit criterion. No password store on legacy parameters; no cryptographic primitive reached through unaudited FFI; every login executes a `verify_and_upgrade` path; every pepper is HSM-resident; PQC migration plan approved at board committee.

Phase 2 — Standardise (months 6–15)

Align internal API contracts with canonical ISO 20022 schemas to ensure data fidelity end-to-end.

Technical work.

- Define a stricter-than-CBPR+ message profile [20, 24] and reject on parse at every internal boundary.
- Translate MT up to MX once at ingress; never carry MT downstream.
- Verify <Dbtr> / <Cdtr> / <DbtrAgt> / <CdtrAgt> carry LEI references end-to-end [47] so sanctions screening becomes auditable rather than heuristic.
- Run dual-running CBPR+ validation against live correspondent traffic for at least one quarter before the November 2026 cut-over to surface field-usage drift.

Organisational change management.

- Stakeholder mapping: Head of Payments, Head of Treasury, Head of Financial Crime, Chief Data Officer; Pillar II sponsor.
- Vendor dependency risk: where a core-banking modernisation programme is in flight (Temenos, Mambu, Finastra, FIS), ensure the strict message profile is enforced at the bank-controlled ingress layer, not at the vendor's — or the bank inherits the vendor's interpretation as policy.
- Staff retraining: payments operations staff retrained on structured-field semantics and exception-handling under reject-on-parse; financial-crime team retrained on LEI-based screening.
- Programme-governance gates: dual-running validation gate at month 12; FATF Recommendation 16 [41] attestation at month 14.

Exit criterion. 95 percent or higher auto-reconciliation rate on cross-border flows; zero MT-only outbound traffic; LEI coverage validated across the top-20 corridors.

Phase 3 — Orchestrate (months 12–24)

Deploy a rail-agnostic control plane that treats SWIFT CBPR+, A2A / PSD3, and tokenised deposits as commodity execution venues governed by policy-as-code.

Technical work.

- Document credit-exposure profiles per rail per corridor.
- Bind the agent to the policy, not the rail.
- Wire SR 11-7 model-risk governance [43] into the orchestration layer (not into the model) and operational-resilience accountability [1, 2, 4–6] above that.
- Implement LCR / NSFR / intraday-liquidity attribution per rail per corridor in line with BCBS 238, BCBS 295, and BCBS 248 [14, 51, 52].
- Treat tokenised deposits and regulated stablecoin rails as stacked, not competing — the corporate sees one ISO 20022 payment regardless of the settlement story underneath.

Organisational change management.

- Stakeholder mapping: Head of Treasury (accountable), Chief Risk Officer, ALCO chair, Head of Correspondent Banking; Pillar III sponsor.
- Vendor dependency risk: tokenised-deposit platforms are not fungible. Onboarding J.P. Morgan Onyx [31] or HSBC Orion [32] is an exposure to that specific platform's governance and reserves model. Document the concentration risk and the exit path before commitment.
- Staff retraining: treasury operations and correspondent-banking teams retrained on multi-rail liquidity attribution; market-risk team retrained on tokenised exposure measurement.
- Programme-governance gates: ALCO sign-off on per-rail credit-exposure profiles; CRO sign-off on the policy-as-code routing rules; internal-audit attestation on the BCBS attribution mappings.

Exit criterion. Every corridor is routable via at least two rails; every rail has a documented credit-exposure profile and a policy-as-code routing rule; every agent action is auditable as a function of policy state, not model state; LCR / NSFR / intraday-liquidity attribution is automated per rail per corridor.

Financial Impact Framing

Compliance argument alone does not move a CFO. The framework's economic case sits on three measurable lines: surcharge avoidance on legacy SWIFT traffic, lift in auto-reconciliation rate, and reduction in cryptographic-control remediation cost under the operational-resilience regimes. *Practitioner ranges, not regulatory floors*. The figures below are illustrative envelopes built from McKinsey's 2024 Global Payments Report cross-border volume baseline [22] and published practitioner observations; institutions should re-anchor them on their own message volumes, average failure-investigation cost, and supervisory expectation.

Line 1 — SWIFT surcharge exposure. The major correspondent banks have published per-message uplifts on translated MT-only traffic effective in the period around the November 2026 cut-over. For a mid-tier transaction bank processing on the order of 2 million cross-border payments per month, a per-message uplift on the order of a few cents compounds to seven-figure annual additional cost before accounting for rejection charges on CBPR+-non-compliant messages. A bank that arrives at the cut-over without an ISO-first ingress carries this line as a recurring drag; a bank that arrives compliant eliminates it.

Line 2 — Auto-reconciliation rate lift. Structured <RmtInf><Strd> [20] carries through correspondent legs without truncation. Baseline auto-reconciliation rates on truncated remittance data are commonly reported in the 80–85 percent range; ISO-first ingress with stricter-than-CBPR+ enforcement is commonly observed to lift the rate into the 92–95 percent range. The economic value of the lift is roughly $V \approx \Delta r \cdot N \cdot c$, where Δr is the reconciliation-rate change, N is annual message volume, and c is the fully loaded investigation cost per failed reconciliation (commonly observed at USD 10–50 in practitioner data, depending on jurisdiction and complexity). For a Tier-1 institution processing N at the low tens of millions of messages per year, a 10-point lift produces an annual operating-cost saving in the high seven to low eight figures.

Line 3 — Cryptographic-control remediation cost. Pillar I substitutes a recurring remediation cost (one-off audits, parameter rotations, FFI dependency reviews) with a substrate cost (memory-safe library maintenance, HSM peppering operations, PQC inventory upkeep). The shift is typically cost-neutral to mildly cost-positive in the first year, then increasingly cost-positive as the recurring audit and incident cost falls. The material economic upside, however, is on the avoided-incident side: under DORA, supervisory fines for severe ICT-resilience failings can reach a percentage of group annual turnover, with the materially larger downside in remediation, customer dispute, and reputational cost.

Composite framing. An ARB-facing financial case typically composes the three lines into an NPV over a three-year horizon, discounted at the institution's cost of capital, with the cryptographic-incident avoidance treated as a tail-risk reduction rather than a steady-state line. The framework is rarely a pure cost-takeout case; it is a tail-risk-reduction plus operating-cost-takeout case, where the relative weight of the two depends on the institution's current incident exposure and ISO 20022 maturity.

Programme-Level Risks

Three programme-level risks deserve standing visibility regardless of phase.

- **Talent and retention.** Memory-safe-cryptography and ISO 20022 deep skills are scarce and mobile. Build internal centres of excellence; do not single-thread the programme through one external consultancy.
- **Regulatory drift.** The five regulatory regimes cited above are not synchronised. Cross-jurisdictional banks need a single internal control framework that maps to each regime's language; let local compliance map back, not forward.
- **Pilot-to-production transition.** Tokenised-deposit and PQC pilots have a long history of stalling at pilot. Set production go/no-go gates at programme start; do not let the pilot phase extend indefinitely.

7 Decision Support: Comparative Posture and Risk Register

The Resilience Trinity is one of several architectural frameworks an Architecture Review Board (ARB) may consider for 2026 modernisation. This section provides explicit comparison against the most-cited alternatives and a programme-level risk register, so ARB members can locate the framework against their existing strategy rather than treating it as a stand-alone proposal.

Comparative Posture Matrix

The four approaches below are not mutually exclusive. Most Tier-1 banks operate elements of two or three concurrently.

(A) Resilience Trinity (this paper). Three modular pillars; ISO-first canonical schema; policy-as-code orchestration outside the agent; pure-substrate cryptographic posture with documented PQC plan. Strength: end-to-end operational-resilience accountability under DORA / SR 21-14 / CPS 230 / MAS TRM / HKMA C-RAF. Weakness: requires internal engineering capacity for the cryptographic and orchestration substrate; not a purchase decision.

(B) Vendor-led core-banking modernisation (Temenos / Mambu / Finastra / FIS). Replace the core ledger and adjacent services with a vendor platform. Strength: ISO 20022 substrate by vendor adoption; operational risk shared. Weakness: cryptographic substrate inherits vendor's choices; orchestration layer is the vendor's interpretation of the bank's policy, not the bank's own policy-as-code; PQC migration depends on the vendor's roadmap; vendor lock-in is structural. *Complementarity.* The vendor handles Pillar II's canonicalisation; the bank still needs Pillars I and III above it.

(C) API-first open-banking platform model. Re-platform the customer-facing layer onto an API gateway and expose payment, account, and balance APIs to corporate clients and fintech partners. Strength: revenue-side velocity; PSD3 / Open Finance alignment. Weakness: leaves the cryptographic substrate and the orchestration question unaddressed; data canonicalisation is at the API contract level, not the message level. *Complementarity.* The API layer sits above Pillar II; the framework's orchestration layer is what makes the API contracts deliverable on a multi-rail estate.

(D) CBDC-rail-led thesis. Build the cross-border product on a future central-bank-digital-currency rail. Strength: settlement risk transferred to central-bank money. Weakness: dependent on CBDC issuance timelines that remain uncertain across most jurisdictions in 2026; does not address the messaging, cryptographic, or orchestration questions on its own. *Complementarity.* A CBDC rail is one corridor option for Pillar III; the orchestration layer makes it routable alongside SWIFT CBPR+ and tokenised deposits.

Programme Risk Register

Top risks identified during framework development. Each is paired with the indicator that signals the risk has materialised and with the mitigating control.

- 1. Cryptographic-asset inventory understates the estate.** *Indicator:* PQC migration plan covers fewer assets than the post-incident inventory used during a tabletop exercise. *Control:* require dependency-graph evidence (SBOM equivalents for cryptographic primitives) at ARB sign-off.
- 2. PQC migration depends on a single HSM vendor.** *Indicator:* no documented fall-back path. *Control:* require dual-vendor demonstrability for any ML-KEM / ML-DSA primitive on the critical path.
- 3. CBPR+ profile is enforced at the vendor edge, not at bank ingress.** *Indicator:* message rejection statistics are vendor-reported, not bank-controlled. *Control:* bank-owned schema-validation layer with rejection telemetry to the bank's own observability stack.
- 4. Tokenised-deposit platform commitment becomes irreversible.** *Indicator:* no documented exit migration of in-flight tokens to alternative rails. *Control:* require exit-path documentation as a precondition of platform onboarding; revisit annually.
- 5. Intraday-liquidity buffer under-states real exposure.** *Indicator:* BCBS 248 daily maximum [14] exceeds 90 percent of the documented buffer on more than five business days per quarter. *Control:* buffer is re-calibrated at quarter end against the previous 90-day peak.
- 6. Policy-as-code rules drift from the model's effective behaviour.** *Indicator:* divergence between the policy decision logged at routing time and the rail actually executed. *Control:* model risk governance under SR 11-7 [43] applied to the orchestration layer, with monthly conformance evidence.
- 7. Regulatory drift creates conflicting attestations.** *Indicator:* the same control evidence is rejected under one regime and accepted under another. *Control:* single internal control framework that maps to each regime's language; local compliance maps back, not forward.
- 8. Skill scarcity stalls the cryptographic substrate.** *Indicator:* memory-safe-cryptography hires take longer than two quarters to source. *Control:* internal centre of excellence; do not single-thread through one external consultancy.
- 9. Pilot extends indefinitely.** *Indicator:* tokenised-deposit pilot exceeds 18 months without a production go/no-go decision. *Control:* ALCO-set production go-live gate at programme start with documented decision criteria.
- 10. Vendor-published PQC roadmap slips relative to NIST IR 8547 [13].** *Indicator:* vendor roadmap misses an interim milestone by more than two quarters. *Control:* contractual right to substitute an alternative supplier on the affected critical-path primitive.

Validation Status and Limitations

The Resilience Trinity is a *synthesis* framework. Each of its three pillars draws on documented practice and published implementations:

- **Pillar I components** (memory-safe cryptographic primitives, multi-algorithm dispatch, HSM-interlocked peppering, PQC migration planning) are individually grounded in NIST, IETF [19], and peer-reviewed cryptographic literature [37, 38].
- **Pillar II components** (an ISO-first canonical schema, stricter-than-CBPR+ profile, reject-on-parse ingress) are documented in the BIS CPMI harmonisation requirements [20] and are operational at multiple Tier-1 banks running the November 2026 dual-running validation period.
- **Pillar III components** (multi-rail orchestration, tokenised-deposit and stablecoin stacking, intraday-liquidity attribution) are exercised by the BIS Project Agorá workstream [8], J.P. Morgan Onyx [31], and HSBC Orion [32].

What is *not* validated is the end-to-end three-pillar model as a single coherent operating posture. No institution has, to the author's knowledge, formally tested or publicly attested to the Resilience Trinity as a unified framework. The cited Tier-1 deployments validate *components*; they do not validate the integration. This is the most material limitation of the paper.

The implication for an ARB considering adoption is twofold. First, the cited risks (Section 7) should be assumed to under-count: an end-to-end framework integration carries integration-specific risks that will surface only in practice. Second, the framework is positioned as an open call to pilot rather than as a settled architecture. Institutions willing to pilot the integration are encouraged to engage; supervisory and academic researchers interested in independent validation are encouraged to take up the question.

What This Section Is Not

This section is not a substitute for an institution's own ARB risk assessment, internal audit attestation, or supervisory dialogue. It is intended to give an ARB member a defensible starting point: a framework located against its alternatives, a risk register against which the institution's own register can be cross-checked, and an explicit acknowledgement of where the framework's empirical evidence ends.

About the Author

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Conflict-of-interest disclosure. The companion technical write-ups cited in Pillars I, II, and III [39, 48, 57] are the author's own published commentary, included for engineering-detail completeness only. The framework's load-bearing references are the standards bodies, regulators, and surveys cited throughout: BIS, NIST, NSA, BCBS, OWASP, IETF (RFC 9106), ENISA, the EU, the Federal Reserve, the OCC, the FFIEC, the FCA, the PRA, MAS, HKMA, APRA, RedCompass Labs, McKinsey, Capgemini, and the BIS Innovation Hub Project Agorá workstream.

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